

Classification: Positive

Text: The furniture is small.

Classification: Neutral

Text: I don't like your attitude

Classification: Negative

Text: That idea was awesome.

Classification:

Chain-of-thought prompting:

There are several uses for chain-of-thought prompting, ranging from arithmetic problem solving to explaining intricate thinking tasks. In educational contexts, where comprehending the process of problem-solving is just as crucial as finding the solution, this is advantageous. It can also be used in programming assistants and customer support bots.

Retrieval-augmented generation (RAG)

RAG was created because LLMs like GPT had limitations. While LLMs have proven to be quite good at producing text, their applicability in the real world is restricted since they often do not produce responses that are appropriate for the context. RAG aims to bridge this gap by offering a system that excels at interpreting user intent and delivering perceptive, context-aware replies.

RAG is essentially a hybrid model that seamlessly blends two significant components. Retrieval-based approach is the process of gathering data from outside knowledge sources, like databases, documents, and web pages. Generative models, on the other hand, are highly adept at producing language that makes sense and is pertinent to the circumstance. What distinguishes RAG is its ability to strike a balance between these two factors, creating a win-win collaboration that allows it to comprehend user enquiries completely and produce accurate, contextually rich answers.

The information capacity constraint of traditional language models is addressed by retrieval augmented generation (RAG). Traditional LLMs

rely on outside knowledge sources to supplement their limited memory, often known as 'parametric memory'. RAG introduces a 'non-parametric' memory. This greatly broadens LLMs' body of knowledge, enabling them to offer more thorough and precise solutions.

Landscape of open source LLMs

Bard, GPT3, GPT4, etc, are closed LLMs that perform exceptionally well; however, since they are not open source, they come with specific limitations.

- **Limited user control:** Closed LLMs developed by large tech companies lack customisation options. Users can't access the underlying architecture or weights of these models, preventing fine-tuning or tailoring them to specific requirements. This limited control hinders innovation and restricts users from fully leveraging the models' capabilities.
- **External service dependency:** To use closed LLMs, users must interact with API endpoints hosted by the respective companies. This dependency on external services introduces latency issues and potential downtime if the service provider experiences outages. Users are also subject to the pricing and availability policies set by the service provider.
- **Data privacy concerns:** When using closed LLMs for inference on sensitive data, users must trust that the data sent to external API endpoints is handled securely. The lack of control over data handling and processing raises privacy risks, especially when dealing with confidential or personal information. Compliance with privacy regulations and data protection laws becomes a concern for users.

On the other hand, open source LLMs offer users transparency,

flexibility, and control, making them an appealing option compared to closed models. These models vary in architecture, number of parameters (impacting performance and hardware needs), datasets used for training, and licences (determining their usage). Open source LLMs provide transparency, flexibility, and customisation options, allowing researchers, developers, and users to advance natural language processing and create innovative language-based solutions.

Nevertheless, open source LLMs are inferior to closed source LLMs, such as GPT-4. This is because closed source LLMs have greater training data and computational capacity.

Let's delve into some prominent open source LLMs and their key attributes.

LLaMA created by Meta: Most leading companies in the LLM market have chosen to develop their LLM in secret. However, Meta has set itself apart. In July 2023, LLaMA 2, a pre-trained generative text model with seven to seventy billion parameters, was released for research and commercial use. Reinforcement learning from human feedback (RLHF) has helped to refine it. It's a generative text model that can be customised for a range of natural language production activities, including programming jobs, and can be utilised as a chatbot. Open, customised versions of LLaMA 2, Llama Chat, and Code Llama have already been made available via Meta.

BLOOM: Launched in 2022, BLOOM is an autoregressive LLM trained to continue text from a prompt on massive quantities of text data utilising industrial-scale computational resources. It was developed over the course of a year-long joint initiative with volunteers from over 70 nations and researchers from Hugging Face.

BLOOM's release was a significant turning point towards the democratisation of generative AI. One of the most potent open source LLMs, BLOOM has 176 billion parameters,